# ForgeDB Project Charter

## 1. Project Objectives

### Project Purpose

ForgeDB is a platform designed to simplify the process of building relational databases from raw data sources such as Excel files, CSV files, and APIs. The system analyzes uploaded data, suggests database structures and relationships, and allows users to review and modify the proposed design before generating a functional PostgreSQL database.

### SMART Objectives

1. Develop a functional MVP within 12 weeks that can import CSV, Excel, and API data and automatically analyze its structure.

2. Provide database schema suggestions, including tables, relationships, and constraints, reducing the manual effort required for database design.

3. Enable users to review, edit, and approve generated database structures before creating a PostgreSQL database.

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## 2. Stakeholders and Team Roles

### Stakeholders

#### Internal Stakeholders

| Stakeholder | Responsibility |
| --- | --- |
| Project Team Members | Design, develop, test, and maintain the system |
| Project Manager | Coordinate project activities and monitor progress |
| Developers | Implement system features and functionality |
| QA Tester | Validate system quality and report issues |

#### External Stakeholders

| Stakeholder | Responsibility |
| --- | --- |
| Holberton School | Academic sponsor and evaluator |
| Instructors and Mentors | Provide guidance and feedback |
| End Users | Use ForgeDB to generate databases from datasets |
| Organizations and Businesses | Potential future users of the platform |

### Team Roles

| Role | Responsibilities |
|---|---|
| Project Manager | Planning, scheduling, task allocation, and progress tracking |
| Team Lead | Supports technical decisions and development coordination |
| Frontend Developer | Develops the user interface and user experience |
| Backend Developer | Develops APIs, authentication, and business logic |
| Data Analysis Developer | Builds the data analysis and schema suggestion engine |
| QA Tester | Performs testing, validation, and defect reporting |
| Documentation Coordinator | Maintains project documentation and reports |

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## 3. Project Scope

### In-Scope

- Upload CSV and Excel datasets.
- Import data from APIs.
- Analyze uploaded data and identify data types.
- Suggest database tables.
- Suggest relationships between tables.
- Suggest database constraints (Primary Key, Foreign Key, Unique, Not Null).
- Allow users to review and modify generated schemas.
- Generate PostgreSQL databases.
- Provide a basic dashboard for database management.

### Out-of-Scope

- NoSQL database generation.
- Real-time data streaming.
- AI model training and predictive analytics.
- Mobile application development.
- Enterprise-scale deployment.
- Advanced database administration features.

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## 4. Project Risks and Mitigation Strategies

| Risk | Impact | Mitigation Strategy |
|---|---|---|
| Limited experience with some technologies | Development delays | Allocate learning time and conduct knowledge-sharing sessions |
| Inaccurate relationship detection | Incorrect schema generation | Allow users to manually review and edit suggestions |

| Risk | Impact | Mitigation Strategy |
|---|---|---|
| Poor quality uploaded data | Reduced analysis accuracy | Implement validation and profiling checks |
| Integration issues between system components | Functional problems | Test each module independently before integration |
| Tight project timeline | Delayed milestones | Prioritize MVP features and follow a structured schedule |
| Unexpected technical challenges | Additional development effort | Maintain flexibility and conduct frequent testing |
| Team communication issues | Reduced productivity | Hold regular meetings and use task-tracking tools |
| Data security concerns | User trust and privacy risks | Implement secure file handling and access controls |

# 5. High-Level Project Plan

## Timeline

| Stage | Duration | Deliverables |
|---|---|---|
| Stage 1: Idea Development | Week 1 | Project idea approved |
| Stage 2: Project Charter Development | Week 2 | Project Charter completed |
| Stage 3: Technical Documentation | Weeks 3-4 | System architecture, database design, and technical specifications |
| Stage 4: MVP Development | Weeks 5-10 | Functional ForgeDB MVP |
| Stage 5: Project Closure | Weeks 11-12 | Final testing, documentation, and presentation |

## Major Milestones

**Milestone 1 – Project Approval**

- Project idea selected and validated.

**Milestone 2 – Project Charter Completion**

- Objectives, scope, stakeholders, risks, and timeline documented.

**Milestone 3 – Technical Documentation Completion**

- Architecture diagrams and technical specifications finalized.

**Milestone 4 – MVP Completion**

- Data upload functionality completed.
- Schema suggestion engine completed.
- PostgreSQL generation functionality completed.

**Milestone 5 – Project Closure**

- Testing completed.
- Documentation finalized.
- Final presentation delivered.

## Simplified Gantt Chart

| Activity | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 | W9 | W10 | W11 | W12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Idea Development | ✓ | | | | | | | | | | | |
| Project Charter | | ✓ | | | | | | | | | | |
| Technical Documentation | | | ✓ | ✓ | | | | | | | | |
| MVP Development | | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | |
| Testing & Bug Fixes | | | | | | | | | ✓ | ✓ | ✓ | |
| Final Presentation & Closure | | | | | | | | | | | ✓ | ✓ |